

Ahmed Saeed (a.k.a. Ahmed Said Mohamed Tawfik Issa)

CONTACT INFORMATION	KACB 3338, 266 Ferst Dr NW Atlanta, GA 30332	asaeed@cc.gatech.edu https://faculty.cc.gatech.edu/~amsmti3/
RESEARCH INTERESTS	Theory, design, and implementation of scalable computer networks and computer systems , including resource scheduling, congestion control, and formal methods.	
EDUCATION	Georgia Institute of Technology , Atlanta, Georgia, USA	
	PhD in Computer Science	August 2019
	Georgia Institute of Technology , Atlanta, Georgia, USA	
	M.Sc. in Computer Science	May 2018
	Alexandria University , Alexandria, Egypt	
	Bachelor's of Science, Computer and Systems Engineering	July 2010
EMPLOYMENT	Georgia Institute of Technology , Atlanta, Georgia, USA	
	<i>Assistant Professor</i>	Aug. 2021 - now
	Massachusetts Institute of Technology , Cambridge, Massachusetts, USA	
	<i>Postdoctoral Associate</i>	Sep. 2019 - Jul. 2021
	Google Inc. , Sunnyvale and Mountain View, CA, USA.	
	<i>Student Researcher</i>	Aug. 2018 - Dec. 2018
	<i>Software Engineering Intern</i>	May 2018 - Aug. 2018
	<i>Software Engineering Intern</i>	May 2016 - Jan. 2017
	<i>Software Engineering Intern</i>	May 2015 - Aug. 2015
	Carnegie Mellon University Qatar and Qatar University , Doha, Qatar.	
	<i>Research Fellow</i>	Nov. 2013 - July 2014
	Egypt-Japan University for Science and Technology , Alexandria, Egypt	
	<i>Research Assistant</i>	July 2010 - Aug. 2013
HONORS AND AWARDS	Nile University , Cairo, Egypt	
	<i>Undergraduate Research Assistant</i>	July 2009 - Aug. 2009
	NSF CAREER Award	
	Faculty Fellow of the Brook Byers Institute for Sustainable Systems (BBISS)	
	DARPA Riser, 2022.	
	Recognized as one of the top five DARPA Risers at the DARPA Forward event, hosted at Georgia Tech.	
	Invited to DARPA Demo Day 2023 at the Pentagon.	
	Class of 1969 Teaching Fellow, 2022.	
	Inclusive STEM Teaching Fellow, 2022.	
	Google PhD Fellowship in Systems and Networking (2017-2019).	
	Third place in Student Research Competition (SRC) at MobiCom 2011.	
	Second place in undergraduate Student Research Competition (SRC) at MobiCom 2009.	

FUNDING

- **CAREER: Integrated Load and Resource Management for High-Utilization Datacenters**, NSF award 2542973
PI, \$595,834, 06/01/2026-05/31/2031
- **NeTS: Small: Understanding and Weathering Variability in Low Earth Orbit Satellite Networks**, NSF award CNS-2345827
PI with Ketan Bhardwaj, \$600,000, 10/2024-10/2027.
- **Collaborative Research: CNS Core: Medium: High-performance Network Stacks for the Edge**, NSF award CNS-2212098
PI with Alex Daglis and Adam Belay (MIT), \$1,198,140 (Georgia Tech: \$748,140) 10/2022-09/2025.
- **Collaborative Research: CNS Core: Medium: Robust Behavioral Analysis and Synthesis of Network Control Protocols Using Formal Verification**, NSF award CNS-2212103
Co-PI with Hari Balakrishnan (MIT), \$1,199,020 (Georgia Tech: \$299,020) 10/2022-09/2025.
- **From Design to Print: An Analytical Study of Circuit Synthesis from Nanomodular Electronics**, Speculative Technologies
PI with Alex Daglis, Unrestricted gift, \$40,000, 08/2023
- **Google Research Award**
Unrestricted gift, \$60,000, 06/2022

CONFERENCE PUBLICATIONS

1. **Svalinn: Overload Control in Large-Scale Servers with Multiple Resource Bottlenecks**
USENIX OSDI 2026
Bhaskar Pardeshi, Peidi Song, Ahmed Saeed
2. **Morp4: A Dynamic Network Telescope**
USENIX NSDI 2026
Iliana Maria Xygykou, Jithin K. Sojan, Dhruv Rauthan, Feng Zhu, Thomas Holterbach, Shane Alcock, Brian Flanagan, Ahmed Saeed, Alberto Dainotti
3. **Governing Datacenter Growth: Regulatory Challenges, Technical Constraints, and Policy Responses Across States**
TPRC 54
Ahmed Saeed, Josiah Hester
4. **Assessing LEO Satellite Networks for National Emergency Failover**
ACM IMC 2025
Vaibhav Bhosale, Ying Zhang, Sameer Kapoor, Robin Kim, Miguel Schlicht, Muskaan Gupta, Ekaterina Tumanova, Zachary S. Bischof, Fabi'an E. Bustamante, Alberto Dainotti, Ahmed Saeed
5. **CoreSync: A Protocol for Joint Core Scheduling and Overload Control of μ s-Scale Tasks**
IEEE ICNP 2025
Bhaskar Pardeshi, Eric Stuhr, Ahmed Saeed
6. **Are LEO Networks the Future of National Emergency Failover? – A Quantitative Study and Policy Blueprint**
TPRC 53
Vaibhav Bhosale, Ying Zhang, Sameer Kapoor, Robin Kim, Miguel Schlicht, Muskaan Gupta, Ekaterina Tumanova, Zachary S. Bischof, Fabi'an E. Bustamante, Alberto Dainotti, Ahmed Saeed

7. **LDB: An Efficient Latency Profiling Tool for Multithreaded Applications**
USENIX NSDI 2024
Inho Cho, Seo Jin Park, Ahmed Saeed, Adam Belay, Mohammad Alizadeh
8. **Protego: Overload Control for Applications with Unpredictable Lock Contention**
USENIX NSDI 2023
Inho Cho, Ahmed Saeed, Seo Jin Park, Mohammad Alizadeh, Adam Belay
9. **A Characterization of Route Variability in LEO Satellite Networks**
PAM 2023
Vaibhav Bhosale, Ahmed Saeed, Ketan Bhardwaj, Ada Gavrilovska
10. **Toward Formally Verifying Congestion Control Behavior**
ACM SIGCOMM 2021
Venkat Arun, Mina Tahmasbi Arashloo, Ahmed Saeed, Mohammad Alizadeh, Hari Balakrishnan
11. **Throughput-Fairness Tradeoffs in Mobility Platforms**
ACM MobiSys 2021
Arjun Balasingam, Karthik Gopalakrishnan, Radhika Mittal, Venkat Arun, Ahmed Saeed, Mohammad Alizadeh, Hamsa Balakrishnan, Hari Balakrishnan
12. **Scouting the Path to a Million-Client Server**
PAM 2021
Yimeng Zhao, Ahmed Saeed, Ellen Zegura, Mostafa Ammar
13. **Annulus: A Dual Congestion Control Loop for Datacenter and WAN Traffic Aggregates**
ACM SIGCOMM 2020
Ahmed Saeed, Varun Gupta, Prateesh Goyal, Milad Sharif, Rong Pan, Mostafa Ammar, Ellen Zegura, Keon Jang, Mohammad Alizadeh, Abdul Kabbani, Amin Vahdat
14. **Overload Control for μ s-scale RPCs with Breakwater**
USENIX OSDI 2020
Inho Cho, Ahmed Saeed, Joshua Fried, Seo Jin Park, Mohammad Alizadeh, Adam Belay
15. **Eiffel: Efficient and Flexible Software Packet Scheduling**
USENIX NSDI 2019
Ahmed Saeed, Yimeng Zhao, Nandita Dukkupati, Ellen Zegura, Mostafa Ammar, Khaled Harras, Amin Vahdat
16. **zD: A Scalable Zero-Drop Network Stack at End Hosts**
ACM CoNEXT2019
Yimeng Zhao, Ahmed Saeed, Ellen Zegura, Mostafa Ammar
17. **Unison: Enabling Content Provider/ISP Collaboration using a vSwitch Abstraction**
IEEE ICNP 2019
Yimeng Zhao, Ahmed Saeed, Mostafa Ammar, Ellen Zegura
18. **Characterizing the Effects of Rapid LTE Deployment: A Data Driven Analysis**
IEEE/IFIP TMA 2019
Kareem Abdullah, Noha Othman Korany, Ayman Khalafallah, Ahmed Saeed, Ayman Gaber
19. **If you can't beat them, augment them: Improving Local WiFi with Only Above-driver Changes**
IEEE ICNP 2018
Ahmed Saeed, Mostafa Ammar, Ellen Zegura, Khaled Harras
20. **Carousel: Scalable Traffic Shaping at End-Hosts**
ACM SIGCOMM 2017
Ahmed Saeed, Nandita Dukkupati, Vytautas Valancius, Vinh The Lam, Carlo Contavalli, Amin Vahdat

21. **Argus: Realistic Target Coverage by Drones**
ACM/IEEE IPSN 2017
Ahmed Saeed, Ahmed Abdelkader, Mouhyemen Khan, Azin Neishaboori,
Khaled Harras, Amr Mohamed
– Full version published in ACM Transactions on Sensor Networks 2019
22. **Local and Low-cost White Space Detection**
IEEE ICDCS 2017
Ahmed Saeed, Khaled Harras, Ellen Zegura, Mostafa Ammar
23. **The Inapproximability of Illuminating Polygons by α -Floodlights**
CCCG 2015
Ahmed Abdelkader, Ahmed Saeed, Khaled Harras, Amr Mohamed
24. **Up and Away: A Visually-Controlled Easy-to-Deploy Wireless UAV Cyber-Physical Testbed**
IEEE WiMob 2014
Ahmed Saeed, Azin Neishaboori, Khaled Harras, Amr Mohamed
25. **Low complexity target coverage heuristics using mobile cameras (*short paper*)**
IEEE MASS 2014
Azin Neishaboori, Ahmed Saeed, Khaled Harras, Amr Mohamed
26. **On Target Coverage in Mobile Visual Sensor Networks**
ACM MOBIWAC 2014
Azin Neishaboori, Ahmed Saeed, Khaled Harras, Amr Mohamed
27. **Location-Aware Probabilistic Route Discovery for Cognitive Radio Networks**
IEEE WCNC 2014
Ahmed Elbagori, Ahmed Saeed, Moustafa Youssef
28. **A Low-Cost Large-Scale Framework for Cognitive Radio Routing Protocols Testing**
IEEE ICC 2013
Ahmed Saeed, Mohamed Ibrahim, Khaled Harras, Moustafa Youssef
29. **RASID: A Robust WLAN Device-free Passive Motion Detection System**
IEEE PerCom 2012
Ahmed E. Kosba, Ahmed Saeed, Moustafa Youssef
– Full version published in IEEE Journal of Selected Topics in Signal Processing 2014

JOURNAL
PUBLICATIONS

1. **On Realistic Target Coverage by Autonomous Drones**
ACM Transactions on Sensor Networks 2019
Ahmed Saeed, Ahmed Abdelkader, Mouhyemen Khan, Azin Neishaboori,
Khaled Harras, Amr Mohamed
2. **Toward Dynamic Real-time Geo-location Databases for TV white Spaces**
IEEE Network 2015
Ahmed Saeed, Mohamed Ibrahim, Khaled Harras, Moustafa Youssef
3. **Ichnaea: A Low-overhead Robust WLAN Device-free Passive Localization System**
IEEE Journal of Selected Topics in Signal Processing 2014
Ahmed Saeed, Ahmed E. Kosba, Moustafa Youssef
4. **Nuzzer: A Large-Scale Device-Free Passive Localization System for Wireless Environments**
IEEE Transactions on Mobile Computing 2013
Moustafa Seifeldin, Ahmed Saeed, Ahmed E. Kosba, Moustafa Youssef, Amr El-Keyi

WORKSHOP
PUBLICATIONS

1. **Modeling the Interactions between Core Allocation and Overload Control in μ s-Scale Network Stacks**
IEEE MASCOTS Workshop
Jehad Hussien, Pratyush Sahu, Eric Stuhr, Ahmed Saeed
2. **Understanding QUIC's Throughput Speedbumps**
ANRW 2025
Saubhik Mukherjee, Demi Lei, Mostafa Ammar, Ahmed Saeed
3. **Lightweight Automated Reasoning for Network Architectures**
HotNets 2024
Rahul Bothra, Akshay Narayan, Venkat Arun, Ahmed Saeed, Brighten Godfrey
4. **Do We Need a Million Satellites in Orbit?**
LEO-NET 2024 in conjunction with ACM MobiCom
Demi Lei, Ahmed Saeed
5. **Astrolabe: Modeling RTT Variability in LEO Networks**
LEO-NET 2023 in conjunction with ACM MobiCom
Vaibhav Bhosale, Ketan Bhardwaj, Ahmed Saeed
6. **Vision: The Case for Symbiosis in the Internet of Things**
ACM MCS 2015 in conjunction with ACM MobiCom
Ahmed Saeed, Mostafa Ammar, Khaled Harras, Ellen Zegura
7. **Towards a Characterization of White Spaces Databases Errors: An Empirical Study**
ACM WINTECH 2014 in conjunction with ACM MobiCom
Ahmed Saeed, Khaled Harras, Moustafa Youssef
8. **Target Coverage Heuristics Using Mobile Cameras**
Robotic Sensor Networks 2014 part of Cyber-Physical Systems Week
Azin Neishaboori, Ahmed Saeed, Amr Mohamed, Khaled Harras

REFEREED
POSTERS AND
DEMOS

1. **Understanding Interactions between Overload Control Core Allocation in Low-Latency Network Stacks**
ACM SIGCOMM 2023 (Poster)
Eric Stuhr, Ahmed Saeed
2. **A Robust Technique for WLAN Device-free Passive Motion Detection**
ACM MobiCom 2011 (Poster)
Ahmed E. Kosba, Ahmed Saeed, Moustafa Youssef
3. **Quantifying the Impact of GPU Specific Optimizations: An Experimental Study on a Weather Forecasting Application**
ACM PACT 2010 (Poster)
Ahmed Saeed, Emad Elwany, Emad Tawadros, Kareem Abdelsalam, Pakinam Yousry, Samia Hafez
4. **DNIS: A Middleware for Dynamic Multiple Network Interface Scheduling**
ACM MobiCom 2009 (Poster)
Ahmed Saeed, Karim Habak, Mahmoud Fouad, Moustafa Youssef
5. **DNIS: Dynamic Multiple Network Interface Scheduler**
ACM MobiCom 2009 (Demo)
Ahmed Saeed, Karim Habak, Mahmoud Fouad, Moustafa Youssef

PATENTS

1. US Patent 10,454,835
“Device And Method For Scalable Traffic Shaping With A Time-Indexed Data Structure”
Carlo Contavalli, Nandita Dukkkipati, Ahmed Issa, Vytautas Valancius, Granted Oct. 2019.
2. US Patent 9,800,946
“Dynamic real-time TV white space awareness”
Khaled Harras, Moustafa Amin Youssef, Mohammed Ibrahim, Ahmed Issa, Granted Oct. 2017.

SERVICE

PC Co-Chair: IFIP Networking 2025.

PC Member:

2027: NSDI
2026: NSDI, CLOUD (Senior PC Member), LEO-NET
2025: CoNEXT, LEO-NET
2024: CoNEXT, INFOCOM, APSys, LEO-NET
2023: NSDI, ANRW, SysDW, LEO-NET
2022: SIGCOMM, HotNets, ANRW, APNet
2021: ANCS

Invited Reviewer IEEE/ACM Transactions on Networking (ToN), IEEE Journal on Selected Areas in Communications (JSAC), IEEE Transactions on Vehicular Technology (TVT), IEEE Transactions on Mobile Computing (TMC), IEEE Computer, IEEE Networking Letters, ACM Transactions on Sensor Networks (TOSN), Elsevier Pervasive and Mobile Computing, Elsevier Ad Hoc Networks, Elsevier Computer Networks, IEEE VTC2015-Spring.

Georgia Tech Service

- SCS Chair Search Committee, 2026.
- Networking Area Coordinator, 2024-2026.
- Member, CoC Communications Faculty Advisory Board, 2022-2025.
- Member, PhD Visit Day Committee, 2022-2024.
- Member, PhD Admissions Committee, 2022.

TALKS

- **Demystifying Datacenters: AI – Energy – Sustainability**
The 38th Red Clay Conference - University of Georgia School of Law Mar. 2026
BBISS Insight Series Feb. 2026
- **Fantastic Congestion Points and Where to Find Them**
Alexandria University Dec. 2024
Applied Innovation Center in Alexandria, Egypt Dec. 2024
American University in Cairo Nov. 2024
Texas Tech University Mar. 2023
- **Improving the Performance and Resilience of LEO Satellite Networks**
DARPA Forward at Georgia Tech Oct. 2022
- **High Performance Network Stacks for Edge Servers**
Google Networking Research Summit 2022 Feb. 2022
- **Building Scalable Network Stacks for Modern Applications**
University of Texas at Austin Apr. 2021
École Polytechnique Fédérale de Lausanne (EPFL) Apr. 2021
University of Toronto Mar. 2021
Microsoft Research Mar. 2021

Imperial College London	Mar. 2021
George Mason University	Mar. 2021
Simon Fraser University	Mar. 2021
University of British Columbia	Mar. 2021
Virginia Tech	Feb. 2021
University of Waterloo	Feb. 2021
Max Planck Institute - SWS	Feb. 2021
Georgia Tech	Feb. 2021
King Abdullah University of Science and Technology (KAUST)	Jan. 2021

- **Scalable Packet Scheduling in Software**

Massachusetts Institute of Technology	Feb. 2019
Princeton University	Feb. 2019
University of Southern California	Oct. 2018
University of California - Santa Cruz	Oct. 2018